

Guardian-Mesh

Enterprise AI Governance & Cost Optimization Layer

Executive Summary: As enterprises scale Generative AI adoption, they face three critical risks: **Uncontrolled Token Costs**, **Data Privacy Leaks (PII)**, and **Regulatory Non-Compliance**. Guardian-Mesh is a high-performance governance proxy designed by Azure Architects to bridge the gap between AI innovation and enterprise security.

Core Value Pillars

 Data Sovereignty Local PII scrubbing using SLMs (Phi-4). Sensitive data is masked 100% locally before hitting public cloud APIs.	 FinOps Intelligence Semantic Caching reduces Azure/AWS token spend by up to 40% by eliminating redundant cloud queries.	 Low-Latency Edge Hybrid routing sends routine tasks to local nodes, achieving sub-millisecond response times for common queries.
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Technical Architecture

- **Universal Gateway:** Cloud-agnostic routing (Azure OpenAI, AWS Bedrock, GCP Vertex).
- **Persistence Layer:** SQLite-backed Audit Ledger for compliance reporting and ROI tracking.
- **Security Mesh:** Real-time regex and NLP-based interception pipeline.
- **Edge Logic:** Local model integration (Ollama/Phi-4) for private inference.

Business Impact & ROI

By deploying Guardian-Mesh, organizations can expect:

- **Immediate Cost Reduction:** 20-50% savings on LLM API bills via Semantic Caching.

- **Security Assurance:** Zero-trust approach to AI prompts, ensuring compliance with GDPR, HIPAA, and CCPA.
- **Audit Readiness:** Automated generation of AI usage logs and cost-avoidance reports.